Amazon Sales Data Analysis

Executive Dashboard Summary — Product Performance, Pricing & Customer Feedback | Prepared for Executive Leadership Review

Executive Summary

This report analyzes 1,351 Amazon products (1,465 listings) across 9 top-level categories, covering pricing, discounting, and 26,766,377 customer ratings. It distills the three-tier dashboard built for this engagement — Basic, Intermediate, and Advanced — into the numbers leadership needs to guide pricing strategy, category investment, and product quality decisions.

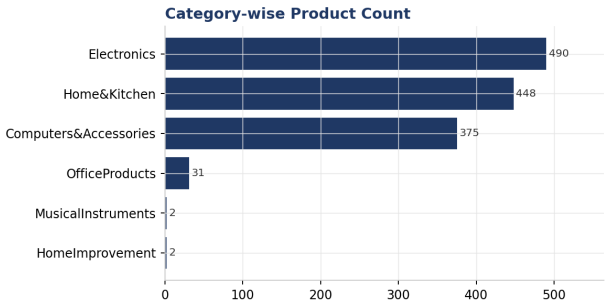
TOTAL PRODUCTS	TOTAL RATINGS	AVG. RATING	AVG. DISCOUNT %
1,351	26,766,377	4.10 / 5	48%

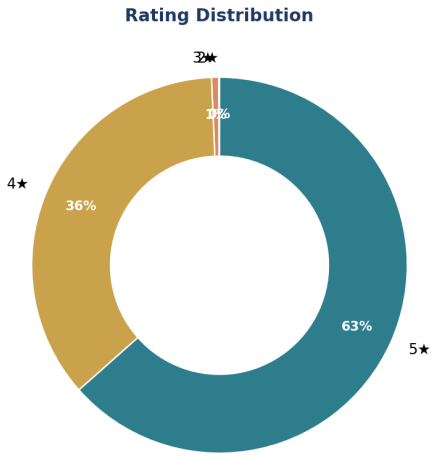
AVG. ACTUAL PRICE	AVG. DISCOUNTED PRICE	TOP CATEGORY (VOLUME)	TOP CATEGORY (ENGAGEMENT)
Rs. 5,445	Rs. 3,125	Electronics	Electronics

Headline signal: Electronics is the company's largest category by product count (490 products) and drives the most customer engagement (15.8M ratings). Products are discounted by 48% on average, cutting the typical price from Rs. 5,445 to Rs. 3,125 — but bigger discounts do not buy better ratings: discount % and rating are essentially uncorrelated (correlation = -0.16, slightly negative), meaning deep discounting is not a reliable lever for improving perceived quality.

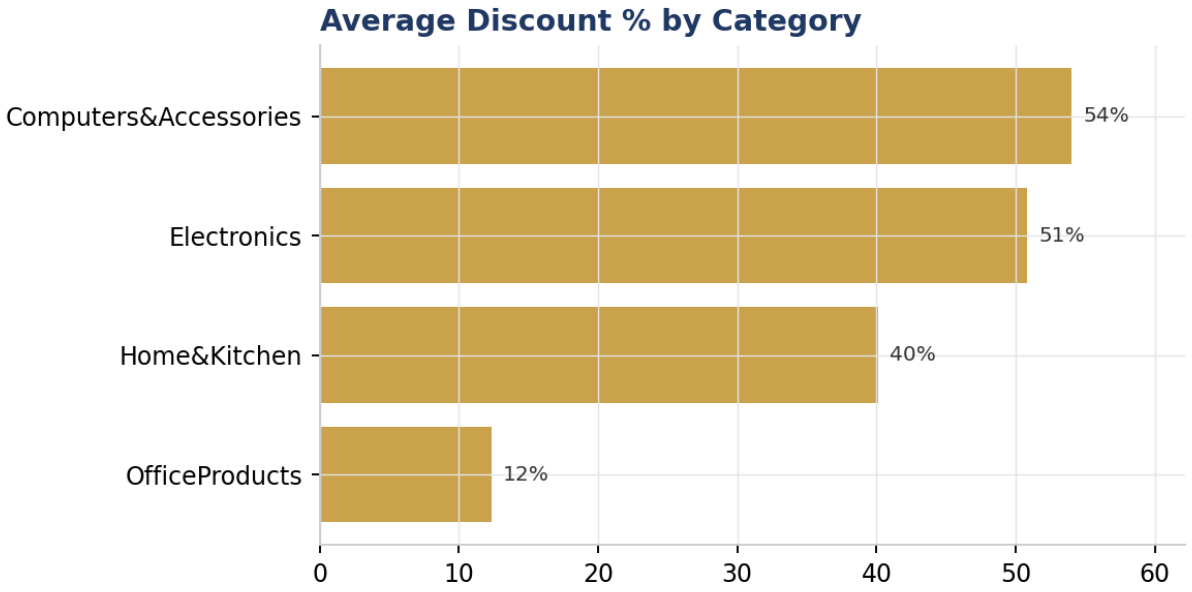
Dashboard 1 — Product Performance & Customer Feedback (Basic)

Slicers on this view: Category, Rating. Visuals: KPI cards, category bar chart, rating donut, top-products table.

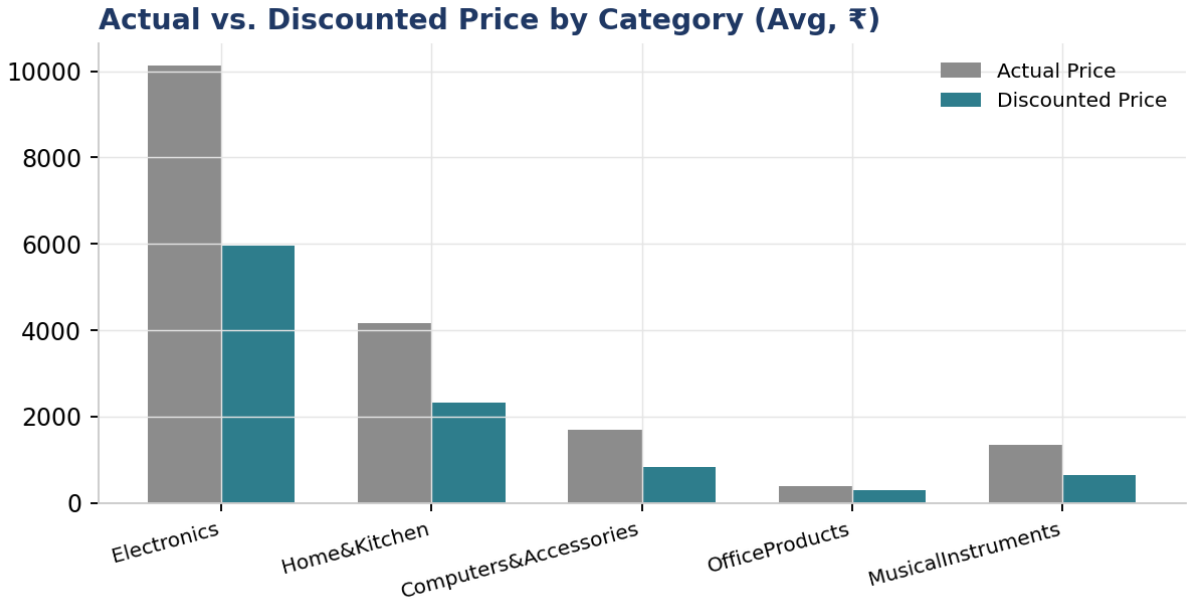




Electronics leads with 490 products. The vast majority of products sit at 4 or 5 stars — ratings below 3 stars are rare in this catalog.



Average Discount % by Category: Computers&Accessories carries the deepest average discount, at 54%, among categories with meaningful product counts.



Actual vs. Discounted Price by Category: every major category sees substantial markdowns from list price, with Electronics showing the largest absolute price gap.

Top 5 Highest-Rated Products (minimum 1,000 ratings)

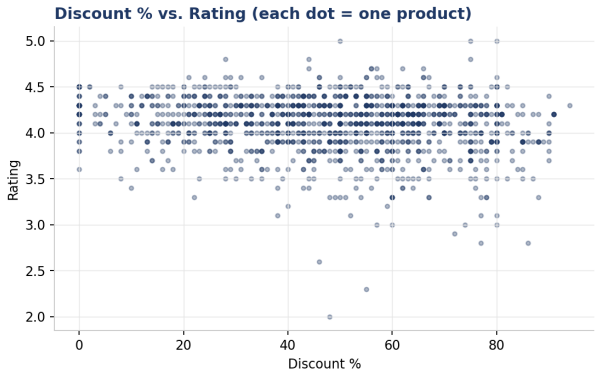
Product	Category	Rating	Rating Count
Swiffer Instant Electric Water Heater Faucet Tap	Home&Kitchen	4.8	53,803
Instant Pot Air Fryer, Vortex 2QT, Touch Control	Home&Kitchen	4.8	3,964
Spigen EZ Fit Tempered Glass Screen Protector fo	Electronics	4.7	7,779
Sony Bravia 164 cm (65 inches) 4K Ultra HD Smart	Electronics	4.7	5,935
Campfire Spring Chef Prolix Instant Portable Wat	Home&Kitchen	4.7	2,591

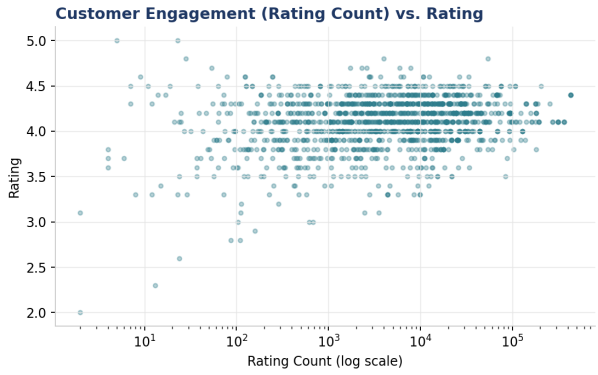
Products with the Highest Discounts (minimum 500 ratings)

Product	Discount %	Rating
rts [2 Pack] Mini USB C Type C Adapter Plug, Type C Female	94%	4.3
Fire-Boltt Ninja Call Pro Plus 1.83" Smart Watch with Blue	91%	4.2
beatXP Kitchen Scale Multipurpose Portable Electronic Digi	90%	3.7
LAPSTER Spiral Charger Spiral Charger Cable Protectors for	90%	4.1
Sounce Protective Case Cover Compatible Boat Xtend Overall	90%	4.0

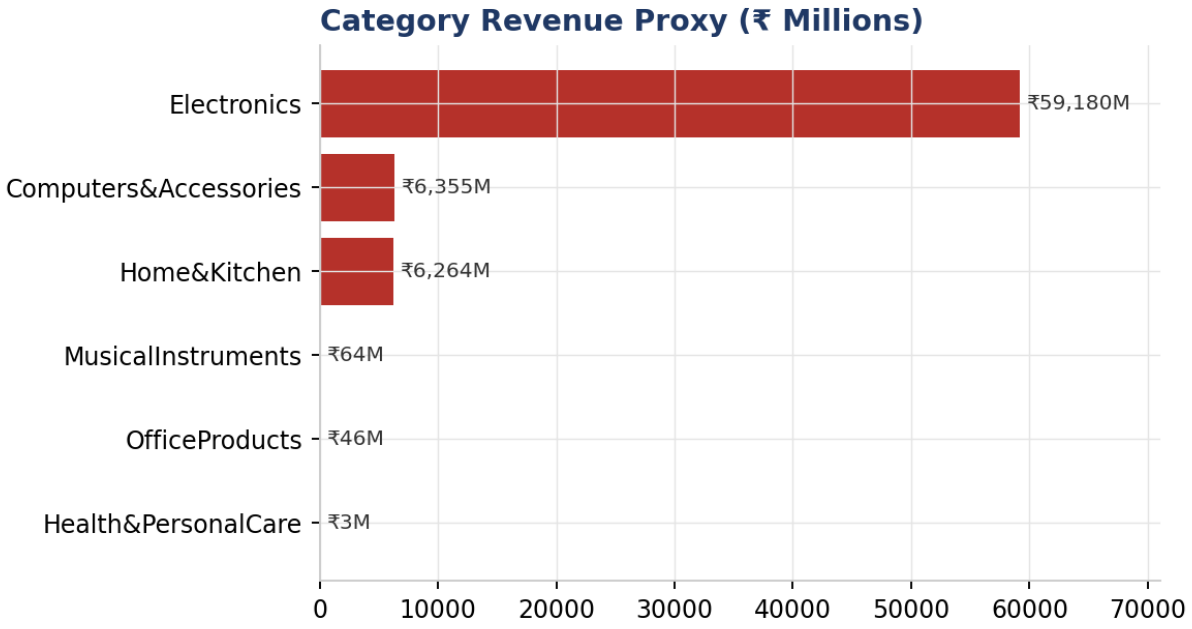
Dashboard 2 — Pricing & Review Insights (Intermediate)

Slicers on this view: Category, Rating. Visuals: scatter plots, category revenue/engagement bars, tooltips on hover showing product-level detail.

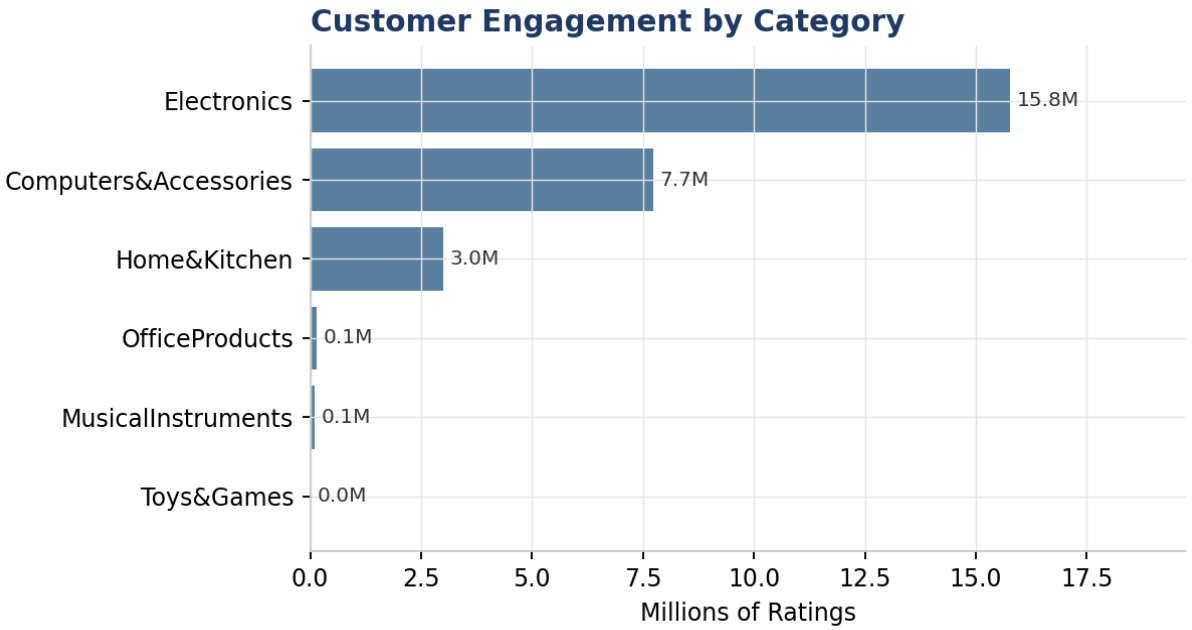




Neither discount depth (correlation = -0.16) nor customer volume (correlation = 0.10) meaningfully predicts a product's star rating — rating is driven mainly by product quality itself, not by price or popularity.



Category Revenue Proxy (rating count × price): Electronics generates by far the largest revenue proxy (Rs. 59,180M), dwarfing every other category — reflecting both its high volume and higher average price point.



Customer Engagement by Category: Electronics accounts for the largest share of total customer ratings, confirming it as the company's highest-visibility category.

Underperforming & Risk Products

- **41 products** have a rating below 3.5 — candidates for quality review or delisting.
- **65 products** combine a steep discount (60%+) with a below-average rating (under 3.8) — discounting alone is not rescuing weak products.
- Review volume trend over time could not be computed: the dataset does not include a review or purchase date field, only aggregated rating counts per product.

Dashboard 3 — Product Scoring & Executive Storytelling (Advanced)

Built using a weighted Performance Score (60% normalized rating + 40% normalized customer engagement) and a dynamic RANKX-style leaderboard, plus discount-optimization and review-mismatch flags.

Top 10 Products by Performance Score

Rank	Product	Category	Rating	Score
1	Swiffer Instant Electric Water Heater	Home&Kitchen	4.8	89
2	Amazon Basics High-Speed HDMI Cable, 6	Electronics	4.4	88
3	AmazonBasics Flexible Premium HDMI Cab	Electronics	4.4	88
4	Amazon Basics High-Speed HDMI Cable, 6	Electronics	4.4	88
5	SanDisk Extreme SD UHS I 64GB Card for	Electronics	4.5	88
6	AmazonBasics USB 2.0 Cable - A-Male to	Computers&Accessories	4.5	85
7	Crucial BX500 240GB 3D NAND SATA 6.35	Computers&Accessories	4.5	85
8	SanDisk Cruzer Blade 32GB USB Flash Dr	Computers&Accessories	4.3	84
9	AmazonBasics USB 2.0 Extension Cable f	Computers&Accessories	4.5	84
10	AmazonBasics USB 2.0 - A-Male to A-Fem	Computers&Accessories	4.5	84

Performance Score = 0.6 × normalized rating + 0.4 × normalized log(rating count), scaled 0–100. Rewards products that are both well-rated and widely purchased/reviewed.

Discount Optimization Opportunities

- **22 products** are top-rated and high-engagement but carry a discount under 30% — strong candidates to test reduced discounting without hurting demand (margin recovery opportunity).
- **3 products** are well-rated but low-visibility with minimal current discount — candidates for a targeted promotional push to build volume on already-good products.

Review–Rating Mismatch Detection

Using a keyword-based text scan of review content (flagging words like "defective," "broken," "waste," "not working"), **64 products** carry a high star rating (4.3+) alongside noticeably negative language in their written reviews — a signal worth a manual quality-team review, since the star rating alone may be overstating actual customer satisfaction for these listings.

Executive Takeaways

- Electronics is the clear priority category — largest catalog, highest customer engagement, and largest revenue-proxy contribution. Any pricing or inventory strategy change here has outsized impact.
- Discounting is not a lever for improving ratings — deeper discounts show a slight negative, not positive, relationship with customer satisfaction. Use discounting for volume/clearance goals, not perceived-quality goals.
- 41 products with sub-3.5 ratings and 65 products stuck in a "high discount, low rating" trap represent the clearest catalog clean-up or supplier-quality opportunities.
- 22 strong performers are under-discounted relative to their demand — a near-term margin recovery opportunity with low execution risk.
- 64 listings show a gap between star rating and review sentiment — worth a manual spot-check, as star ratings alone may not be fully reliable for these products.

Appendix — Power BI Build Reference

- Data model: single fact table (products) with Category split into a hierarchy (Main Category → sub-categories) via Power Query text split on "|".
- Key DAX: Total Products = DISTINCTCOUNT(product_id); Total Reviews = SUM(rating_count); Avg Rating = AVERAGE(rating); Revenue Proxy = SUMX(Products, rating_count * discounted_price).
- Ranking: Performance Score and Rank = RANKX(ALL(Products), [Performance Score], , DESC) for a dynamic, slicer-aware leaderboard.
- Recommended visuals: KPI cards, category bar charts, rating donut, scatter plots (discount vs. rating; rating count vs. rating), decomposition tree on category → rating, tooltip page with product-level detail.
- Slicers: Category, Rating (1–5), Discount % range.
- Note: no transaction date field exists in this dataset, so a true time-series revenue or review-volume trend cannot be built without an additional dated sales/orders table.